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Fairtrade in Peru: Challenges and Opportunities for Sustainable and Equitable Agricultural Development

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Abstract: Fairtrade, present in Peru since 1990, has been widely studied for its impact on agri-food production. This research, based on convenience surveys conducted with leaders and members of cooperatives and agri-exporting companies during the 2024 Peruvian Expoalimentaria fair, examines the benefits and challenges faced by producers in their pursuit of sustainable and equitable development. The results were statistically analysed using Cronbach's alpha, factor analysis, and the ordinal logistic regression model. These methods revealed that not all the dimensions evaluated reached statistical significance. However, the dimensions of agricultural sustainability and social justice emerged as the most significant, driven by the adoption of sustainable technologies and democratic participation in cooperatives. Nonetheless, these dimensions require additional external conditions to ensure well-being, such as market incentives, which did not achieve statistical significance, suggesting the influence of contextual factors not accounted for in the study. This research not only enriches the existing literature but also provides recommendations for implementing Fairtrade initiatives in Peru to promote more equitable and sustainable agricultural development that improves the socio-economic and socio-political conditions of the most disadvantaged producers. Furthermore, this study highlights the need to expand the theoretical development of unexplored variables, such as association models, sustainable practices, and economic growth in rural communities. In this way, it serves as an exploratory foundation for future public policies and strategies in the agri-food sector.

Keywords: Fairtrade; market incentives; sustainable agriculture; social justice; agricultural development; Peru; development sustainability

check for updates

Academic Editor: Jacob Arie Jordaan

Received: 28 November 2024
Revised: 3 January 2025
Accepted: 4 January 2025
Published: 10 January 2025

Citation: Lozano Paredes, D.L.; Okada Saavedra, H.; Moscoso Cuaresma, J.R.; Azabache Moran, C.A.; Yesquén Delgado, K.N.d.P.; Diaz Cruz, M.E.; Salazar Seminario, V.L.; Pastor Pinto, J.; Amer Layseca, T. Fairtrade in Peru: Challenges and Opportunities for Sustainable and Equitable Agricultural Development. *Sustainability* **2025**, *17*, 486. https://doi.org/10.3390/su17020486

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1. Introduction

Since the 1940s, Fair Trade initiatives have grown, seeking to promote sustainable production standards and improving the conditions of socio-economically and environmentally disadvantaged small-scale producers [1]. As a result, Fairtrade provides the living conditions of agricultural producers, ensuring fair prices and equitable working conditions. Certification promotes the welfare of producers in developing countries and ensures that high standards are met through pricing and certification strategies [2]. Additionally, Fairtrade benefits workers and sets high standards for production, delivery, working conditions, and socio-environmental policies in supply chains, contributing to sustainable development and protecting small producers from industrial exploitation [2].

First, Fairtrade organisations promote fairer trade by supporting producers and raising public awareness, with the aim of reforming conventional international trade rules [3]. The literature indicates that the benefits of Fairtrade are not always evenly distributed. Fairtrade relies on existing social and cultural capital, mainly certifying already-organised operators to access global markets [4]. Large exporting companies can offer higher prices to producers than Fairtrade cooperatives, thanks to their economies of scale, the use of futures markets, and increased funding [5]. Although farmers manage to become certified, many are unable to sell all their production under Fairtrade standards, which restricts the economic benefits they could gain due to price volatility [6].

In the Peruvian context, Luna [7] mentions that the agri-food export sector has played a crucial role in the economy, generating approximately 800,000 direct and indirect jobs in recent years, leading Latin America in 2020 in terms of obtaining Fairtrade certifications in the coffee sector [8]. All this has led to improved integration in international markets, positioning itself as a leader in the global market [9]. One of the challenges for certified producers is to access markets that demand their products, which makes the intervention of intermediaries key to facilitating the conditions for filling the market gap [4]. As recommended by Guevara-Rosero et al. [10], the intermediary should not only guide small producers through the certification process but also support them in selling their products, which directly impacts their income levels and contributes to more equitable economic opportunities.

The purpose of Fairtrade is to "improve the plight of the world's poor and disadvantaged producers" and "ensure development" by changing the terms of trade [11]. Its influence on agri-food production is reflected in market incentives, the promotion of sustainable agriculture, and the improvement of social justice. Certification guarantees that companies comply with Fairtrade standards, ensuring the integrity of the system [12]. Consumers, by valuing ethical aspects in their purchases, drive Fairtrade, promoting equitable development and encouraging shared responsibility between producers, companies, and consumers in a fairer and more sustainable system [13].

To deepen this understanding, a visit will be made to Expoalimentaria 2024, where we plan to collect quantitative information that will help to better understand the impact of Fairtrade on producer communities. This fair is a key event in the agri-food sector in Latin America, offering a platform for interaction between producers, entrepreneurs, buyers, and experts in the sector [14]. This event brings together exhibitors from all over the world, specialising in fresh and processed foods, agricultural inputs, machinery, and technologies for the food industry. Our participation allowed us to directly contact Fairtrade-certified producers and subsequently survey cooperatives and companies regarding the implementation of sustainable practices.

There are gaps in the regulation of Fairtrade that affect small-scale agri-food producers. Although certifications seek to improve their conditions, they do not guarantee that the benefits reach those who need them most, generating uncertain results [10]. A key problem is that, although certifications are costly, there are no clear rules to limit the negative impact of these costs on producers, who bear the economic burden without support from certifiers or governments to balance the situation [13]. Furthermore, the lack of adequate oversight allows large agri-food corporations to maintain unequal relations with small producers, damaging the organisations of the latter years within the Fairtrade system [15]. In short, the lack of a solid legal framework to regulate certification and trade relations in Fairtrade leaves small producers in a vulnerable situation, without adequate protection from the system. Moreover, underdeveloped commercial practices in Latin American and Asian countries exhibit systemic patterns rooted in historical inequalities, limited access to resources, and dependence on global markets [16,17]. These practices are often characterised

by unequal value distribution across supply chains, where rural producers face challenges such as low wages, lack of infrastructure, and limited market opportunities [18]. The literature highlights that addressing these issues requires a multi-faceted approach that balances economic equity with environmental sustainability. Efforts to prioritise sustainable practices, equitable income distribution, and environmental protection aim to empower rural communities by fostering self-reliance, improving market access, and encouraging long-term resilience in agricultural production

The objective of this study is to analyse the influence of Fairtrade on agri-food production, focusing on three key aspects: market incentives, agricultural sustainability, and social justice. It seeks to assess how Fairtrade impacts Peruvian small-scale farmers by analysing their access to equitable markets, the promotion of sustainable agricultural practices, and socio-economic improvements. Through a detailed analysis, it will identify how these factors contribute to rural development, poverty alleviation, and the reduction of inequalities in the Peruvian agricultural context. The results will provide a clear view on the current effects of Fairtrade in Peru, serving as a basis for the design of future policies and strategies that promote greater equity and sustainability in global value chains, contributing to a fairer and more efficient model for all actors involved.

2. Theoretical Framework

2.1. Fairtrade

Fairtrade emerged in 1988 as a strategy to improve the economic and social conditions of small producers in developing countries through a non-profit initiative in the Netherlands that ensured fair remuneration for low-income farmers [19]. Since then, this global movement has sought to reduce poverty, promote transparency in supply chains, and establish fairer trade relations [20]. It is based on ten key principles, including the creation of opportunities for disadvantaged producers, transparency in supply chains, and environmental protection through sustainable practices [10].

Fairtrade emerged as a solidarity movement with the aim of establishing alliances between organisations and activists in developed countries and small producers in developing regions [19,21]. This movement not only seeks to improve farmers' working conditions, but also to secure fair prices for their products, encourage sustainable agricultural practices, and promote Fairtrade relations [22]. Furthermore, Fairtrade certification has enabled small producers in Latin America to increase their sales and income due to slightly higher prices [19]. In addition, Guevara-Rosero et al. [10] confirmed that certified companies experience a positive economic impact compared to non-certified ones.

Despite the benefits, the economic impact is not always favourable, especially for small farmers. The high cost of certification excludes the most vulnerable producers, and minimum prices, which are not adjusted to the market, can lead to competitive disadvantage [2]. In addition, benefits tend to be concentrated on better-off producers, exacerbating inequality within the value chain [19]. This phenomenon of inequality has already been pointed out, showing that Fairtrade organisations in Mexico, especially in the coffee sector, are vulnerable to the strategies of multinationals, leaving the most disadvantaged exposed to the challenges of the market [23].

In the Peruvian context, Fairtrade has had a significant impact, particularly in the agri-export sector, by enabling small-scale farmers to access international markets with fairer prices and equitable trading conditions [1,24]. This approach has been crucial for products such as coffee, cocoa, and fruits, where Fairtrade standards have allowed rural producers, many of whom are in economically disadvantaged areas, to improve their quality of life through guaranteed prices and development premiums. However, as in other Latin American countries, challenges remain [17]. The high cost of certification,

combined with the pressure from value chains dominated by large multinationals, limits the participation of the most vulnerable producers. This phenomenon reflects the structural inequalities that Fairtrade seeks to address, highlighting the need to strengthen policies of inclusion and sustainability within this system.

2.2. Market Incentives

Market incentives in Fairtrade motivate producers and companies to adopt ethical and sustainable practices, generating a global competitive advantage through mechanisms that encompass economic, social, and environmental aspects. According to Valkila et al. [6], one of its key objectives is to support small farmers' cooperatives with guaranteed minimum prices, premiums for social development, and long-term trading relationships, improving labour rights and economic stability. In addition, Ribeiro-Duthie et al. [25] highlight that it encourages sustainable practices such as avoiding pesticides and training farmers. Dragusanu et al. [19] stress the importance of long-lasting relationships that improve working conditions and facilitate access to finance. Ruben et al. [2] point to key principles such as transparency, gender equity, and environmental protection. Finally, Barro-Chale et al. [1] state that while promoting sustainable practices and improving food security in the Peruvian rainforest, it faces challenges such as lack of fair payment, although it still offers social justice and global competitive advantage.

Fairtrade market incentives emerged with certification to increase sales of labelled products in mainstream retail outlets [11]. Producers also reduce their exposure to price fluctuations, mitigate risks, and improve investment [26]. This movement connects farmers in the Global South with consumers in the North, improving production and working conditions, especially in sectors such as coffee [5]. Fairtrade also offers price premiums to farmers who adopt organic practices, supporting them during long conversion periods [27]. Since its inception, Fairtrade has improved the position of small-scale producers by granting market access, eliminating middlemen, and ensuring fair prices, according to research on the origins and development of this movement [28].

Fairtrade market incentives include guaranteed minimum prices and social and organic premiums, which have outperformed market prices over the last decade, providing crucial income for vulnerable producers [11]. In Uganda, Fairtrade certification has improved the living standards of smallholder coffee farmers by 30% and reduced poverty [29]. In addition, it facilitates credit for producers without access to other sources of finance [5]. However, not all aspects of Fairtrade are always beneficial, since Kurjanska and Risse [30,31] suggest that while it may appear to be a sustainable development strategy, in some cases, it is not. Furthermore, according to Alberto [31], although the principle of solidarity applies, not all aspects of Fairtrade are completely virtuous, facing challenges that limit its positive impact in some contexts.

Fairtrade market incentives have improved the living conditions of small producers, offering minimum prices and premiums that strengthen their economic stability. They promote environmental and social sustainability, encouraging responsible practices and skills development in globalised markets. However, there are limitations, such as the difficulty of achieving fair payment and criticisms about the fulfilment of their sustainable development promises. Despite these challenges, their impact remains relevant to improving conditions for marginalised producers.

In conclusion, Fairtrade initiatives have played a critical role in strengthening the position of small-scale producers, particularly in regions like the Peruvian rainforest and the Andean highlands. By offering guaranteed minimum prices and social premiums, these initiatives have provided a safety net for vulnerable agricultural communities, enabling them to invest in sustainable practices and improve their economic stability. Furthermore,

the promotion of organic farming methods, as highlighted by Ribeiro-Duthie et al. [25], aligns with Peru's rich biodiversity, fostering environmentally friendly practices and reducing dependency on harmful chemicals. The focus on gender equity and long-term trading relationships, as noted by Ruben et al. [2], is particularly impactful in empowering female farmers and promoting inclusive economic growth in traditionally underserved areas. However, challenges such as the inconsistency in fair payments and limited market access remain barriers that must be addressed to fully realise the potential of Fairtrade in Peru. Despite the obstacles, Fairtrade initiatives have contributed significantly to enhancing the global competitiveness of Peruvian agri-exports, particularly in sectors like coffee and cacao, by meeting international standards for ethical and sustainable production [1].

2.3. Social Justice

The term "social justice" was first mentioned in the mid-19th century, focusing on a fundamental equality between individuals, without linking it specifically to the labour or economic sphere [32]. It was not until a century later, however, that this term acquired a new meaning in the context of labour and industrial relations. The International Labour Organisation (ILO) broadened this concept, linking it to labour rights, as this factor is crucial to guarantee universal and lasting peace, by linking it directly to the promotion of internationally recognised human and labour rights. This pioneering approach laid the foundations for the evolution of the concept in social and political thinking. In Fairtrade, it advocates fairness and respect, with principles such as the prohibition of child labour, gender equality, and decent working conditions [33].

The literature mentions that Fairtrade has played a key role in promoting the development of countries in the Global South by offering fairer trading conditions and guaranteeing labour rights [20]. It is also noted that female participation in production has increased, promoting gender equality in agricultural cooperatives [34]. In addition, gender training initiatives and women's associations have been implemented in rural areas where male migration or the demand for female participation is relevant [35]. On the other hand, the integration of public and private regulations has strengthened social and environmental sustainability in supply chains, creating fairer value chains [36]. Fairtrade not only generates economic benefits, but also strengthens social cohesion, promotes inclusion, and fosters sustainable development that responds to the needs of marginalised producers.

However, there are studies to the contrary, highlighting the importance of further analysis on effectiveness in promoting social justice [28]. It has been shown that, in some cases, workers' wages in certified cooperatives were lower than those in non-certified cooperatives, because of the costs associated with certification [10]. Furthermore, it is noted that, although certified farmers can obtain higher prices, in rural regions workers continue to receive only the minimum wage and lack essential benefits [19]. This evidence underlines the need to re-evaluate the Fairtrade system to ensure that its benefits favour all members of the supply chain, especially the most vulnerable, and not just large cooperatives or middlemen [37].

Fairtrade, although it has made significant progress, is limited in its ability to solve the problems of the most marginalised and must address more dimensions of poverty and inequality [38].

2.4. Sustainable Agriculture

Sustainable agriculture can be defined as a system that seeks to balance agricultural production with the preservation of natural resources, contributing to the economic and social well-being of rural communities [39]. This approach is based on the implementation of agricultural practices that respect the environment, such as integrated pest management

and organic agriculture [40]. Although Fairtrade does not always have a direct environmental impact, its standards encourage improvements in farming practices that benefit the sustainability of the ecosystem [41]. On the other hand, consumers are willing to pay a premium price for certified products, reflecting a growing support for sustainability in agriculture [42].

Organic agriculture and Fairtrade offer access to stable markets, institutional prices and contribute to economic sustainability by reducing dependency on agrochemicals. Fairtrade not only guarantees better incomes but also encourages the conservation of natural resources through responsible agricultural practices, such as the protection of water sources and the reduction of chemical fertilisers [39], In Costa Rica, farmers participating in sustainable agriculture programmes have managed to maintain between 30% and 50% shade cover on their coffee plantations, which promotes biodiversity and soil conservation [43]. Furthermore, the choice of sustainable materials, such as sacks and organic fertilisers, together with the predominant use of pack animals as a means of transport, highlights the commitment to sustainable practices on the part of farmers [44].

Critics point out that the creation of local organic markets in developing countries is a challenge, as farmers face difficulties in accessing markets due to low demand and competition with conventional products [45]. Fairtrade certification, while promising better incomes, does not always translate into the full adoption of sustainable environmental practices, as producers often continue to rely on cash crops that do not meet environmental standards [39]. In addition, the high costs of Fairtrade and organic certifications are a significant barrier for smallholder farmers, who face difficulties in covering these costs without immediate benefits, limiting their access to these markets [46]. Finally, organic farming is more labour-intensive compared to conventional systems, which increases operating costs, especially in areas where labour shortages are a problem [44].

The evidence on barriers to the creation of local markets for sustainable products underlines the importance of Fairtrade initiatives to improve market access for small-scale producers. This allows an assessment of whether Fairtrade programmes can mitigate these challenges in Peru. The transition to sustainable practices, without adequate economic incentives, suggests that Fairtrade can offer the necessary financial support to Peruvian farmers [39]. The costs of certification and extra labour in organic agriculture require assessing whether the economic benefits of Fairtrade outweigh these barriers, strengthening agri-food sustainability [47–49].

2.5. Agri-Food Production

Agri-food production encompasses all activities that transform agricultural resources into food for human and animal consumption, from traditional small-scale methods, which have existed for centuries, to more modern intensive systems [50]. However, intensification, driven by the need for food and energy, has had negative consequences such as soil degradation, loss of biodiversity, and pollution of water resources, as well as aggravating poverty in the most vulnerable regions [51,52]. Although intensive agribusiness has prioritised increasing production at low cost to maximise profits, traditional agricultural practices continue to play an important role in food security in many regions [50,53].

In this context, regenerative agriculture is based on natural practices that seek to restore and strengthen the resources used in production, imitating nature's own processes to ensure sustainability [52]. Similarly, Fairtrade has provided small-scale farmers with access to international markets, improving working conditions and food security, and generating a positive economic impact on rural communities [22]. Agroforestry, by combining trees with crops or livestock, increases agricultural productivity, improves biodiversity and soil health, and contributes to greater climate resilience. This system also diversifies farmers' incomes,

offering additional products such as timber and fruit [50]. Environmental benefits, such as improved water retention and reduced erosion, also make agroforestry a key solution for long-term agricultural sustainability [54].

Agricultural techniques that degrade soil fertility, decrease carbon content, and reduce the amount of living things in ecosystems are called "degenerative" [52]. This affects their long-term productive capacity, decreasing biodiversity and causing erosion, which compromises the sustainability of agricultural systems [55]. In addition, the excessive use of pesticides and fertilisers in these systems has affected key ecosystems such as pollinators, putting crop stability and environmental services at risk [56]. Intensive agricultural practices have also led to the pollution of water bodies, as chemical fertiliser residues leach into rivers and lakes, also harming human health [54]. On a social level, intensive agriculture perpetuates precarious working conditions for workers in many developing countries, with low wages and few labour protections, keeping these workers in cycles of poverty [22].

Agricultural intensification has caused negative impacts, but practices such as regenerative agriculture and agroforestry restore ecosystems. Fairtrade supports these efforts, improving labour conditions, market access, and rural economic well-being.

2.6. The Development of Fairtrade in Peru

The impact of Fairtrade on Peruvian agri-exporters has been widely documented, particularly in strategic sectors such as coffee, cocoa, and quinoa. Evidence suggests that Fairtrade initiatives play a crucial role in improving limited opportunities in the agri-food sector [1]. These initiatives significantly promote socio-economic and ecological development through responsible, sustainable, and ethical practices, supported by awareness-raising and certification of partner producers, as well as challenges to deepen the political impact of a movement based on social justice and environmental sustainability [57]. They also generate positive spill-over effects, such as improved access to credit, capital strengthening, increased investments, and changes in attitudes towards risk [2,24].

However, some studies conclude that there are no significant differences between Fairtrade and conventional trade chains in terms of improving producers' quality of life [58]. Even with the use of technologies that, from an ethical and transparency perspective, seek to position Fairtrade in international markets [17], the institutionalisation of these practices faces significant challenges. In Peru, Fairtrade is considered fragile and is limited by the producers' strong dependence on value chains, as well as by the lack of organisational dynamics that promote learning and autonomy, fundamental elements for improving producers' quality of life [59].

These points are widely debated in the literature. While it is acknowledged that inequalities and exploitation may exist in Fairtrade supply chains, these tend to affect the most marginalised actors, such as local producers and seasonal workers, who are often invisible in research [59,60]. Despite these divergences, the literature agrees that Fairtrade certification in Peru contributes to the growth and development of small agricultural producers, giving them recognition in international markets. This certification provides the buyer with the guarantee of acquiring quality products that comply with social, economic, and environmental standards [61,62].

2.7. Fairtrade in Latin America

Fairtrade in Latin America, especially in countries with similar cultural and idiosyncratic realities to those of Peru, Colombia, Ecuador, and Bolivia, presents common patterns in its impact and challenges [63]. In all these contexts, Fairtrade certifications have preferentially boosted access to international markets and promoted sustainable agricultural practices [17], in strategic sectors such as coffee, cocoa, quinoa, and bananas [38]. However,

there is a continued dependence of small producers on international value chains, which limits their autonomy and capacity for development [58], even though producers promote Fairtrade, sustainable practices, and responsible consumption, connecting nearly 2 billion participants to global markets worth USD 8 billion annually [11]. Moreover, while these initiatives improve marketing conditions [40] and generate international awareness, the benefits do not always reach the most vulnerable actors, such as seasonal labourers and smallholder farmers, who are often invisible in research and excluded from direct economic benefits [15,21].

Despite these challenges, there are significant differences in implementation and results between these countries. For example, Ecuador has managed to strengthen its cooperatives, maximising the impact of Fairtrade in sectors such as bananas and cocoa [10]. Colombia, for its part, has made progress in the articulation between the public and private sectors, which has improved access to credit for agricultural producers [6]. In contrast, Peru and Bolivia face greater difficulties in forming solid organisational structures and adapting to international regulations, which limits the sustainability of Fairtrade initiatives [23]. In all cases, there is a perceived need for more robust public policies that effectively support these practices.

In summary, although Fairtrade has proved to be a valuable tool for socio-economic and sustainable development in Latin America, its real impact depends on structural and organisational factors [57]. The region faces common challenges, such as the lack of autonomy of producers and inequality in value chains, but also shows differentiated opportunities depending on the local context. To maximise benefits, it is essential to strengthen organisational dynamics, promote producer empowerment, and ensure that the benefits of Fairtrade reach all actors in the supply chain, especially the most marginalised [17].

2.8. Hypothesis Development

The implementation of Fairtrade initiatives in Peru promotes sustainable agricultural practices and social justice in rural communities through the adoption of sustainable technologies and democratic participation in cooperatives, although market incentives have limited influence due to unconsidered contextual factors. In this context, Fairtrade acts as a mediating factor that facilitates access to markets, promotes sustainable practices, and improves the working conditions of producers [4,28]. Furthermore, the certifications associated with Fairtrade generate trust and transparency, in line with a growing environmental and social awareness, which not only reinforces the reputation of companies, but also opens new markets for their ethical and sustainable products [24,42,57].

This hypothesis is related to arguments previously documented in the literature on the impact of Fairtrade on Peruvian agri-exporters, especially in strategic sectors such as coffee, cocoa, and quinoa [1,2,24,58]. Fairtrade initiatives have proven to be crucial in improving the limited opportunities of the agricultural sector, promoting socio-economic and ecological development through responsible, sustainable, and ethical practices [28]. These initiatives also generate positive side effects [59], such as improved access to credit, strengthened capital, increased investment, and changes in attitudes towards risk [2,24]. However, significant challenges remain, such as the institutionalisation of these practices, the fragility of Fairtrade value chains, and the dependence of producers, elements that limit their autonomy and learning [60,61]. Despite the criticisms, Fairtrade certification in Peru continues to be recognised for contributing to the development and growth of small agricultural producers, giving them visibility in international markets and assuring consumers of products that meet social, economic, and environmental standards [1]. Accordingly for these arguments, the following hypotheses were developed:

H1. *The relationship between market incentives and Fairtrade-mediated agri-food production is positive.*

The international market incentive positively influences Peruvian Fairtrade agri-food production by motivating producers to improve the quality and sustainability of their products, with the aim of accessing more lucrative markets and satisfying the growing demand for value-added products. This incentive drives the adoption of improved operational practices that promote more sustainable production [24], reduce environmental impact, and optimise the use of natural resources [17,24,39,58]. Furthermore, Fairtrade initiatives act as a catalyst in this process, fostering not only economic and environmental sustainability, but also strengthening social justice through certification, access to international markets and organisational support to small producers [59,60]. However, the effectiveness of these practices may vary according to local structural and organisational contexts, highlighting the importance of addressing challenges related to dependence on value chains and lack of producer autonomy to maximise the positive impact of Fairtrade.

H2. *The relationship between sustainable agriculture and Fairtrade-mediated agri-food production is positive.*

Sustainable agriculture seeks to minimise the environmental impacts of agri-food production in Peru [24], while Fairtrade promotes equitable practices for producers [58]. In other words, Fairtrade facilitates the adoption of sustainable techniques by acting as a mediator between these practices and the results in Peruvian agri-food production. This approach not only strengthens the efficiency of the agricultural sector, but also boosts its sustainability, promoting a balance between economic, social, and environmental development. Furthermore, by encouraging the implementation of sustainable technologies and ethical practices, Fairtrade contributes to consolidate more inclusive and responsible supply chains, reinforcing the resilience of the agricultural sector in the face of global challenges, as has been widely explained in the literature [4,17,59,60].

H3. *The relationship between social justice and Peruvian agri-food production mediated by Fairtrade is significant and positive.*

The relationship between social justice and Peruvian agri-food production mediated by Fairtrade has been significantly linked to the dynamics of benefit distribution in this model. Although Fairtrade is based on principles of equity, research by Mook et al. [4,28] and Guevara-Rosero et al. [10] has pointed to partial results, highlighting limitations in the improvement of working conditions and in the equitable distribution of the income generated. Although progress has been made in increasing prices for producers, challenges remain in improving the quality of life of the most vulnerable actors within these value chains. These findings underline the relevance of social justice [24] as a key component in optimising the impact of Fairtrade in Peruvian agri-food production [17,58], highlighting the need to re-evaluate certification mechanisms to guarantee more inclusive and sustainable benefits [59].

3. Methodology

This study uses a quantitative approach to analyse the impact of Fairtrade certification on the agri-food production of Peruvian companies, focusing on three components: market incentives, sustainable agriculture, and social justice. The sample includes 43 Fairtrade-certified companies, which were surveyed during the Expoalimentaria 2024 fair.

On the other hand, advanced statistical tools, such as factor analysis and ordered logistic regression, were used to assess firms' perceptions of the effects of certification on the components. The survey questions are based on previous research, which ensures their relevance and accuracy.

For the market incentives (MI) block, we used the article developed by Dragusanu et al. [19] and Ibnu et al. [64] which highlight the economic benefits and long-term relationships that Fairtrade offers to producers. Regarding the sustainable agriculture (SA) block, the references used were from Barro-Chale et al. [2] and Makita [39] which link certification to sustainable practices aligned with the Sustainable Development Goals. Finally, the social justice (SJ) block draws on the work of Mook et al. [4] which addresses the social benefits of Fairtrade, such as improved working conditions and the prevention of child labour.

On the other hand, to achieve the objectives of the study, the conceptual model presented in Figure 1 of the article is proposed as a split section to provide a clear structure to guide the research. This approach is based on a comprehensive review of the existing literature on Fairtrade and its dimensions.

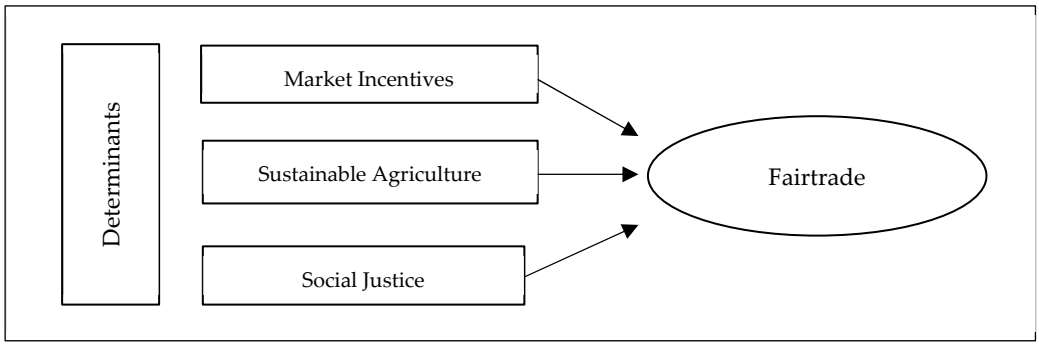


Figure 1. Conceptual model of variables and dimensions. Note. Own elaboration.

The conceptual framework presented in Figure 1 was based on a comprehensive review of the literature on the proposed benefits of Fairtrade and used theoretically in the work developed by Mook and Overdevest [28]. In addition, three main dimensions were identified: market incentives, sustainable agriculture, and social justice. The market incentives dimension, which was developed in Section 2.2, was based on research into how Fairtrade functions as a mechanism to provide producers in developing countries with advantageous access to consumer markets. The social justice dimension was based on studies that have analysed the impacts of this certification in terms of empowerment of marginalised farmers, especially women, decision-making, and improved quality of life, and which were theoretically discussed in Section 2.3. Sustainable agriculture was based on work that has explored how Fairtrade promotes more environmentally friendly agricultural practices among certified producers and was extensively discussed in Section 2.4.

This multidimensional conceptual framework allowed for a comprehensive assessment of the various impacts of Fairtrade, encompassing economic, social, and environmental aspects. By structuring the analysis around these three key dimensions, the evidence sought to provide a solid basis for examining the effectiveness of the Fairtrade system in meeting its 10 stated principles. Similarly, this conceptual model also drew on literature examining producers' expectations and satisfactions with respect to these proposed benefits of Fairtrade [24,27,28]. This approach made it possible to assess whether the Fairtrade certification system is delivering on its promises from the producers' perspective.

The conceptual model of the study integrated these three components to determine the impact of Fairtrade on Peruvian agri-food companies, and they are conceptually explained in Figure 1.

In developing the survey, thematic scientific evidence was consulted and was useful in formulating the key items. For example, in the market incentives (MI) block, item MI1 was based on the findings of Ibnu et al. [64], who stated that certified farmers received

increased economic benefits and market opportunities, as well as training that fostered capacity building and strengthened the sense of community.

Similarly, item MI2 was based on the work developed by Dragusanu et al. [19], who argued that Fairtrade not only guarantees better prices for producers, but also fosters long-term relationships with buyers, thus improving bargaining conditions. These questions reflect the objective of assessing how Fairtrade influences companies' access to more competitive international markets, which is essential to measure companies' perceived eco-economic incentives.

In the sustainable agriculture (SA) block, the questions were also linked to existing literature. For example, item SA3 highlighted that Fairtrade promotes the sustainable use of natural resources as a fundamental strategy to reduce poverty and improve social justice [39]. Furthermore, item SA6 was based on research developed by Guevara-Rose et al. [10], who point out that Fairtrade is a social trend that drives sustainability through practices that align with the Sustainable Development Goals.

In terms of the social justice (SJ) block, item SJ1 was based on the work developed by Mook and Overdevest [28], who underline that one of the key objectives of Fairtrade is to improve working conditions, such as preventing child labour and promoting safe conditions for workers.

Each item developed was supported by scientific evidence. The Market Incentives (MI) dimension was supported by work developed by Konuk [13], Dragusanu et al. [19], Guevara-Rosero et al. [10], Raynolds et al. [11], Mook and Overdevest [4], and Konuk [13].

The Sustainable Agriculture (SA) dimension was developed under the research of the following authors: Chávez Becker and Jurado Celis [15], Makita [39], Barro-Chale et al. [1], Schaafsma et al. [65], Guevara-Rosero et al. [14,35], and Lyon [35].

Finally, the Social Justice (SJ) dimension was based on the work of Mook and Overdevest [4,28], Valkila and Nygren [5]. For more details on the questionnaire, please see Table 1.

Table 1. Questionnaire by item.

Dimensions	Item	Author	Questions
Market Incentives (MI)	MI1	[13]	Revenues from certification.
	MI2	[19]	Price bargaining power.
	MI3	[10]	Access to customers in international markets.
	MI4	[11]	Competitiveness as a result of certification.
	MI5	[19]	Demand for certified products.
	MI6	[4]	Productivity as a result of supply chain improvement.
	MI7	[13]	Economic incentives for access to international markets.
Sustainable Agriculture (SA)	SA1	[15]	Acquisition of new sustainable technologies.
	SA2	[39]	Natural resource management.
	SA3	[39]	Implementation of sustainable agricultural practices.
	SA4	[1]	Sustainable agricultural resources.
	SA5	[65]	Improved sustainable production methods.
	SA6	[10]	Competitiveness due to sustainable agricultural practices.
	SA7	[35]	Democratic participation in decision-making.
Social Justice (SJ)	SJ1	[4]	Prevention of child labour.
	SJ2	[4]	Commitment to sustainable agriculture.
	SJ3	[28]	Inclusion of traditionally marginalised groups.
	SJ4	[5]	Quality of life of workers.

Note. Own elaboration.

3.1. On the Use of Principal Component Analysis (PCA) and Ordinal Logistic Regression (OLS)

To validate that the questionnaire items effectively represented the three theoretical dimensions proposed, principal component analysis (PCA) was used according to the

theory proposed by Mook and Overdevest [28]. This statistical approach allowed a complex set of relationships to be reduced to a smaller number of components, facilitating the interpretation of the data. The PCA provided a solid basis for examining whether the Fairtrade certification system is delivering on its promises from the producers' perspective. By structuring the analysis around these three dimensions, it sought to provide a comprehensive assessment of producers' perceptions of the benefits of Fairtrade, encompassing economic, environmental, and social aspects.

Additionally, the selection of ordinal logistic regression (OLS) was based on the literature that determines its ability to examine multivariate relationships and determine which factors have the greatest influence on producers' overall satisfaction with the Fairtrade certification system [28]. This approach made it possible to assess how different aspects of the programme contributed to farmers' overall perceptions, providing valuable information on which elements are most crucial to its success as perceived by participants.

3.2. Procedure

The methodology used was based on data collection through a structured survey using the Google Forms digital form tool, applied during Expoalimentaria 2024, on 25, 26, and 27 September in Lima, Peru. For this purpose, a closed questionnaire was designed using a five-point Likert scale. This scale made it possible to measure the intensity of respondents' perceptions of the key variables under study.

Before starting the survey, informed consent was sought from all participants, ensuring compliance with the ethical principles of confidentiality.

3.3. Population and Sample

In the present study, non-probabilistic convenience sampling was used to survey 43 companies certified under Fairtrade standards in Peru. Due to the vastness of Peru, it was decided to carry out the surveys at Expoalimentaria 2024, one of the most important events in the sector, which brought together national and international Fairtrade-certified companies. Peru has more than 300 organisations linked to this practice, mainly cooperatives and associations of small producers in key sectors such as coffee, cocoa, and bananas [66].

This approach was chosen mainly for logistical and timing reasons, as this event concentrates a significant representation of relevant companies, thus facilitating access to data [13,67,68]. In addition, Expoalimentaria 2024 brings together both national and international certified companies, providing a suitable environment to explore perceptions on the impacts of Fairtrade certification.

On the other hand, convenience sampling, while practical and efficient, has inherent limitations. In this case, the results reflected the perceptions of the companies participating in Expoalimentaria 2024, but not necessarily those of all Fairtrade-certified companies in Peru. However, it is important to highlight that León [68] mentions 5% of Peruvian agri-exports in a year are negotiated at Expoalimentaria 2024. Likewise, this fair generated business expectations of USD 695 million, an increase of 28.7% over the previous year, consolidating itself as a key platform for connecting small and medium Peruvian producers with high-level international buyers from Europe, Asia, North America, and other regions, thus facilitating the promotion of innovative and sustainable products in line with the objectives of Fairtrade [69]. This reflects its importance as a platform to connect small producers and Fairtrade-certified companies with buyers from international markets with greater purchasing power, thus fulfilling one of the fundamental objectives of Fairtrade: to facilitate access to trade opportunities that promote more equitable and sustainable conditions.

In terms of sample size, 33 certified companies and 44 individuals operating under Fairtrade policies were surveyed during the Expoalimentaria 2024 fair. This number

of companies represented approximately 10.64% of the estimated total of 310 Fairtrade-certified organisations in Peru, according to data from Agraria [66]. This sample size was considered sufficient for the exploratory objectives of the study, as it allowed identifying key trends and assessing the perceptions of company representatives regarding the impacts of certification on the components analysed: market incentives, sustainable agriculture, and social justice; however, the results of this survey need to be taken with caution as, since convenience sampling was applied, the sample may not be fully representative.

The convenience sample was conducted on the following grounds: (i) companies that are Fairtrade-certified, complying with international standards, such as Fairtrade or the World Fairtrade Organization (WFTO); and (ii) companies that have international reach. It is worth noting that, in 2020, Peru led the world in receiving the Fairtrade Premium, accumulating a total of EUR 33,887,687 which reflects its commitment to and leadership in Fairtrade practices [57]. For more details on the distribution of companies, the reader is recommended to see Table 2.

Table 2. Participation by company.

		n	Percentage
Companies	Agro. Sol de Olmos	2	5%
	Alto Aruya	1	2%
	Aprocaf Valle del Monzón	1	2%
	Aso. de Prod. Bio Emprende.	1	2%
	C.A. Ecovraem	1	2%
	Café aicasa	1	2%
	Coop A. Bagua Grande	1	2%
	Coop A. de Jóvenes Latda	1	2%
	Coop A. Industrial Atalaya	1	2%
	Coop A. Usuarios Rio y Valle	1	2%
	Coop. A. Agroex. huangala	1	2%
	Coop. A. Aprosaroch Mazamari	1	2%
	Coop. A. de Serv. Mult. Cordi. Azul	1	2%
	Coop. A. Frutos del Ande	2	5%
	Coop. A. Villa Rica	1	2%
	Coop. Apropal	1	2%
	Coopapbosf	1	2%
	Cooperativa Agraria Acuicola Campovida	5	12%
	Cooperativa Agraria Cafetalera Divisoria	1	2%
	Cooperativa Agraria Industrial Asproc	1	2%
	Cooperativa Agraria Land Perú	1	2%
	Cooperativa Agroindustrial Yarinacocha	2	5%
	Don Augusto	1	2%
	Ecoandino	2	5%
	Finka Kivinaki	1	2%
	Kulkao	1	2%
	Laboratorio de Productos Naturales Takiwasi	1	2%
	Pangoa Cooperativa Agraria Cafetalera	1	2%
	Proassa	2	5%
	Proginger	1	2%
	San Juan del Oro	1	2%
	Solkfrut del Campo	2	5%

Note. The number of Fairtrade-certified companies present at Expoalimentaria 2024 is shown. It also reflects the importance that companies attach to complying with Fairtrade standards, which guarantees fairer conditions for producers. Therefore, this result highlights the growing adoption of this label in the agri-food sector.

On the other hand, the territorial data of the participating companies are detailed in Table 3. This analysis allows us to identify the regions with the highest concentration, such as Piura and Junín, as well as those with the lowest representation, providing a clear picture of the geographical dispersion of the companies included in the study.

Table 3. Distribution of companies by department n = 43.

Departments	n	Percentage
Amazonas	1	2%
Ayacucho	2	5%
Chiclayo	2	5%
Cusco	1	2%
Huánuco	3	7%
Junín	11	26%
Loreto	2	5%
Piura	12	28%
Puno	1	2%
San Martin	4	9%
Ucayali	4	9%

Note. This was performed to show the distribution of the 43 companies analysed according to departments.

Table 4 was developed to analyse the types of organisations involved in the study, identifying the predominance of small producers' cooperatives, as well as the participation of private companies and hired labour organisations. This analysis is fundamental to understanding organisational diversity within the context assessed.

Table 4. Types of organisations.

Types of Organisations	n	Percentage
Small producers' cooperative	27	62%
Private company	6	14%
Organisation of contract workers	10	24%

Note: The range of organisations participating in Expoalimentaria 2024 is illustrated.

Despite having a convenience sample, the approximate geographical segmentation based on the companies surveyed at Expoalimentaria 2024 shows representation with the national distribution of Fairtrade-certified organisations, validating the relevance of the sample [70] and taking into consideration the geographical distribution proposed by Mincetur [71].

Given that there is no proportionality of Fairtrade-certified companies in Peru, as evidenced by the report developed by Mincetur [71], it was decided to analyse the certified companies and participants of Expoalimentaria 2024 at an aggregate level instead of carrying out a regional analysis.

For example, there is a predominance of companies surveyed in the north, central, and jungle regions, and in contrast, a lower representation of the south region, reflecting the structural and logistical limitations faced by this area, also in line with its lower proportion of Fairtrade certifications. These findings highlight that the sample surveyed is not only representative of the companies participating in Expoalimentaria 2024, but also reflects the disparity in the proportion of companies from the southern region at the national level.

Additionally, an analysis by product was carried out as shown in Table 5, with coffee and cocoa standing out as the most representative Fairtrade-certified products in Peru. Coffee shows a remarkable distribution in the four macro-regions, with greater relevance in the central region, while cocoa stands out in the centre and the jungle. This behaviour reflects its alignment with international market demands, where both products enjoy a consolidated position due to their quality and sustainability, which are also essential factors in the preferences of global consumers.

It is worth noting that the Peruvian government has implemented programmes to strengthen coffee and cocoa production, focusing on improving their quality, sustainability, and attractiveness in international markets [72]. These initiatives include technical training,

access to modern technology, and active promotion at international events. Therefore, this institutional support has been key to increasing the competitiveness of these crops, and to gain more visibility in the international market.

Table 5. Main exported products.

Products	n	Percentage
Ginger	5	11%
Aguaymanto and Mango	6	13%
Cocoa	10	22%
Banana	6	13%
Coffee	14	30%
Cotton and Alpaca	1	2%
Quinoa	2	4%
Palm oil	1	2%
Palm Heart Preserves	1	2%

Note. The main Fairtrade-certified products are presented, highlighting those that are produced under Fairtrade conditions.

However, it is important to consider the bias inherent in the data, as the data only reflect information from companies that participated in Expoalimentaria 2024, excluding those that did not attend. Despite this limitation, the results show clear trends that highlight the strategic importance of products such as coffee and cocoa in Peruvian exports [67].

The leading products include coffee, cocoa, and fruits, underlining the relevance of these categories in international trade and sustainable practices (see Table 5).

The productive capacity of the companies is reflected, which gives an insight into the scale of operations under the Fairtrade model. The graph highlights the differences between larger companies and smaller cooperatives (see Table 6).

Table 6. Annual production of the surveyed companies.

		n	Percentage
Production	Between 100 and 500 tonnes	16	38%
	More than 500 tonnes	18	40%
	Less than 100 tonnes	9	21%

Note. Annual production volumes in tonnes are shown for the companies surveyed at Expoalimentaria 2024.

A high percentage of the production of Fairtrade-certified companies is destined for international markets, which is evidence of the global demand for products produced under Fairtrade standards (see Table 7).

Table 7. Percentage of production destined abroad.

		n	Percentage
Overseas production	Between 25% and 50%.	4	10%
	More than 50	35	81%
	Less than 25	4	10%

Note. Percentage of annual production exported abroad is shown.

3.4. Distribution of the Main Peruvian Fairtrade Products by Macro-Region

The geographic segmentation of the surveyed companies at the Expoalimentaria 2024 fair demonstrates representativeness consistent with the national distribution of Fairtrade-certified organisations, validating the relevance of the sample. The central and northern regions, which lead in Fairtrade certifications nationwide with 27.47% and 38.46%, respectively [70], showed proportional participation in the conducted surveys, with the central region (30.3%) and the northern region (18.2%) standing out at the fair. This pattern,

also observed in the jungle and southern regions, reinforces the validity of the results obtained and their alignment with nationally documented dynamics, according to the methodology established by Mincetur [71].

The predominance of the northern and central regions in the surveys and Fairtrade certifications can be attributed to structural and economic factors favouring their agricultural competitiveness, such as irrigation projects driving technological advancements in the north and the central region's strategic location near major logistical and commercial hubs in the country [71,73]. In contrast, the lower representation of the southern and jungle regions in the sample reflects the structural and logistical challenges these areas face, which also align with their smaller proportion of Fairtrade certifications. These findings highlight that the surveyed sample is not only representative of the companies participating in Expoalimentaria but also accurately captures the regional proportions of participation in Fairtrade practices at the national level (see Table 8).

Table 8. Main exported products by macro-region.

Macroregion	Main Exported Products		
	Product	**n**	**%**
Centre	Total Centre	33	39.76%
	Cocoa	8	9.64%
	Coffee	7	8.43%
	Camu-camu	2	2.41%
	Ginger	1	1.20%
	Native fruit juices	1	1.20%
	Quinoa	2	2.41%
	Oats	2	2.41%
	Chia	2	2.41%
	Turmeric	1	1.20%
	Beans	2	2.41%
	Maca	2	2.41%
	Purple corn	2	2.41%
	Oranges	1	1.20%
North	Total North	20	24.10%
	Physalis (Goldenberry)	6	7.23%
	Coffee	2	2.41%
	Bananas	6	7.23%
	Mango	6	7.23%
Jungle	Total Jungle	28	33.73%
	Cocoa	5	6.02%
	Coffee	2	2.41%
	Camu-camu	4	4.82%
	Clavo huasca	1	1.20%
	Palm heart preserves	1	1.20%
	Ginger	4	4.82%
	Bananas	2	2.41%
	Basil	1	1.20%
	Blueberries	2	2.41%
	Turmeric	2	2.41%
	Oregano	1	1.20%
	Golden Pineapple	2	2.41%
	Vanilla	1	1.20%
South	Total South	2	2.41%
	Coffee	2	2.41%

4. Results

4.1. Application of Likert Scale and Cronbach's Alpha with SPSS Tool

Using the analysis in SPSS V.29 software, Cronbach's Alpha was calculated for three key dimensions of the study: market incentives (MI), sustainable agriculture (SA), and social justice (SJ) (see Table 9), where a quantitative approach is used to analyse the impact of Fairtrade certification on the agri-food production of Peruvian companies. The methodology included several statistical analyses to ensure the validity and reliability of the results; however, only significant results were considered $\geq$0.3 in each bold component, excluding from the analysis items MI5, MI5, M7 in the market incentives dimension, as well as items SA2, SA4, SA6 and SA7 in the sustainable agriculture dimension, and item SJ4 in the social justice dimension.

Table 9. Assessment of key factors for business sustainability using reliability analysis and factor loadings.

Item.	Reliability Statistics	Factor Loadings				
	Market Incentives (α = 0.751)	**1**	**2**	**3**	**4**	**5**
MI1	Revenues from certification	**0.301**	−0.029	−0.066	−0.008	0.087
MI2	Price bargaining power	**0.315**	−0.080	−0.034	0.026	−0.049
MI3	Access to customers in international markets	−0.017	0.104	−0.005	**0.382**	−0.142
MI4	Competitiveness following certification	−0.057	−0.141	0.080	**0.512**	0.233
	Sustainable agriculture (α = 0.501)					
SA1	Acquisition of new sustainable technologies	**0.349**	−0.121	0.116	−0.235	−0.150
SA3	Implementation of sustainable agricultural practices	−0.009	−0.045	**0.402**	−0.094	−0.015
SA5	Improved sustainable production methods	−0.060	−0.089	−0.067	0.078	**0.692**
	Social justice (α = 0.630)					
SJ1	Prevention of child labour	−0.115	**0.390**	0.003	0.091	−0.232
SJ2	Commitment to sustainable agriculture	−0.025	**0.335**	−0.143	−0.021	0.174
SJ3	Inclusion of traditionally marginalised groups	−0.112	−0.039	**0.468**	0.187	−0.157
	Improving the quality of life of workers	−0.032	**0.306**	0.106	−0.134	−0.042

Note. The degree of significance of the item with its respective dimensions is shown.

To begin the study process, a survey was carried out at the Expoalimentaria 2024 fair using a Likert scale, a tool used in this type of research to measure people's opinions or attitudes on a specific topic. It consists of a series of statements in which respondents indicate whether they agree or disagree with a situation, generally on a scale of 1 to 5, where 1 is totally disagree and 5 is totally agree.

This was followed by an exploratory factor analysis (EFA), which is a statistical technique that allows for simplifying and more accurately understanding a complex data set. Its main objective is to identify underlying structures or factors that explain the correlations between multiple observed variables. According to Mavru [74], the AFE seeks to establish an internal structure between the variables analysed, generating new factors from a set of variables, especially when there is no prior knowledge of this structure. As a result of this analysis, the Cronbach's alpha reliability index could be determined according to a given group, assessing the internal consistency of the questionnaire in its different blocks (market incentives, sustainable agriculture, and social justice). This indicator ensured that the items within each block consistently measured respondents' perceptions, considering an acceptable value above 0.7.

Finally, an ordered logistic regression model was used to analyse and predict outcomes that are organised into categories with a logistic order. With this method, it was also possible to analyse which item was most relevant to the research.

It is also important to mention that this article was mainly based on the basic paper by Mook and Overdevest [28], which ensures its relevance.

Cronbach's alpha analysis shows that the market incentives dimension has good internal consistency, indicating that the items are well related and reliably measure the construct ($\alpha = 0.751$). On the other hand, the sustainable agriculture dimension ($\alpha = 0.501$) reflects low reliability, suggesting that the items are not sufficiently correlated or may be addressing heterogeneous aspects, which requires revision to improve their consistency. Finally, the social justice dimension ($\alpha = 0.630$) shows moderate consistency, although still insufficient to be considered ideal, which could be due to the diversity of the items included, such as item SJ1 and SJ4.

The results presented in this table allow us to identify five groups among the respondents whose views and interests differ from each other. Firstly, it was observed that the first group agrees that MI1 and MI2 are considerably relevant for an improvement of agri-food production. Similarly, item SA1 is also considered to have a positive impact on the variable in question. This is understood in this way because when analysing the results, they were greater than (0.3) which indicates the significance of this item for the variable in question. Secondly, there is a group particularly interested in social justice aspects, such as SJ1, SJ2, and SJ4. This leads to the understanding that, for this group, there is a greater impact on agri-food production if these items are met. Thirdly, there are two relevant aspects for this group, namely SA2 and SJ3. These aspects are highly considered due to their scores, which were higher than the scores for SA3 and SJ3 (>0.4). Fourthly, there is again a group interested in market incentives; this includes the items MI3 and MI4. It is observed that the last-mentioned item, i.e., competitiveness, is an outstanding fact, as it has a significance level of (0.512). Finally, in the last group detected, a particular situation was found which, for the respondents, was of utmost importance for better development of agri-food production. With a coefficient of (0.692), the highest value obtained, this is the implementation of better sustainable methods.

4.2. Factor Analysis (FA)

Factor analysis in SPSS is used to examine surveys for common factors that explain the behaviour of firms in specific contexts, such as Fairtrade in the agri-export sector. This technique simplifies the business decision-making process by highlighting the key dimensions that influence the success of these practices.

The analysis allowed the original variables to be grouped into five main groups, each capturing different aspects of the variance in the data, in which, factor loading coefficients greater than 0.3 are considered significant in the component matrix.

First, in component 1, three significant coefficients were found, which relate to economic benefits and the adoption of sustainable technology driven by Fairtrade certifications. Then, in component 2, three other significant coefficients were identified, linked to social and labour impact. Then, in component 3, two significant coefficients were found related to agricultural sustainability and access to international markets, resulting from the sustainable agricultural practices adopted. On the other hand, in component 4, two other significant coefficients were identified, associated with improved competitiveness and expansion in international markets. Finally, in component 5, a significant coefficient was found, related to the adoption of sustainable methods in agricultural production.

Moreover, Figure 2 and Table 10 show each breakpoint with a steeply sloping line, representing the group of respondents whose perspectives are similar (see precisely).

Thus, the figure represents the number of components found in the factor analysis (of five components according to Table 10) such that each segment identified by a point represents a component identified at 70.214% of the total variance explained by the factor regression.

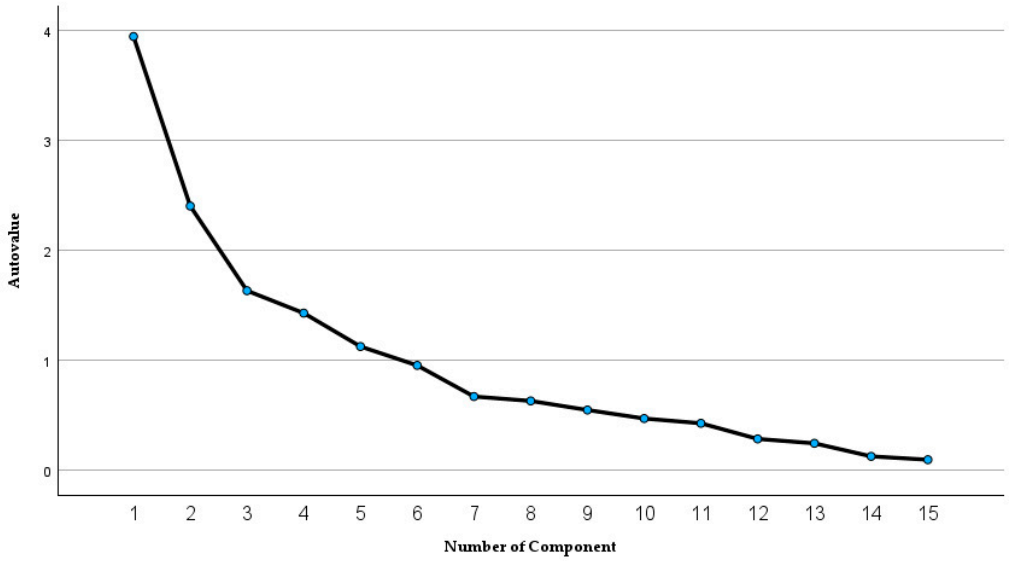


Figure 2. Sedimentation chart of factor analysis. Note. Own elaboration in SPSS V.29.

Table 10. Total variance explained of factor analysis.

Component	Initial Eigenvalues			Sums of Squared Extraction Charges			Sums of Charges Squared by Rotation		
	Total	% Variance	% Accumulated	Total	% Variance	% Accumulated	Total	% Variance	% Accumulated
1	3.942	26.278	26.278	3.942	26.278	26.278	2.809	18.730	18.730
2	2.402	16.012	42.290	2.402	16.012	42.290	2.441	16.272	35.002
3	1.633	10.887	53.177	1.633	10.887	53.177	2.070	13.800	48.802
4	1.429	9.530	62.707	1.429	9.530	62.707	1.785	11.899	60.701
5	1.126	7.507	70.214	1.126	7.507	70.214	1.427	9.513	70.214
6	0.954	6.363	76.576						
7	0.672	4.478	81.054						
8	0.632	4.211	85.266						
9	0.549	3.663	88.929						
10	0.472	3.147	92.076						
11	0.429	2.858	94.933						
12	0.287	1.912	96.845						
13	0.247	1.646	98.492						
14	0.128	0.855	99.347						
15	0.098	0.653	100.000						

Extraction method: principal component analysis.

Note. Own elaboration in SPSS V.29.

4.3. Ordinal Logistic Regression (OLS)

4.3.1. Market Incentives

It has not been considered to place the table regarding market Incentives. This is because the results indicate that the variables analysed do not have a statistically significant impact on the dependent variable in this model, which may suggest that other external or internal factors not considered may be influencing the results ($p < 0.05$) on the dependent variable in this model, which may suggest that other external or internal factors not considered may be influencing the results.

4.3.2. Sustainable Agriculture

The parameters that are significant for the research are those related to the acquisition of new technologies and the use of sustainable agricultural resources ($p < 0.05$) (see Table 11).

Table 11. Parameter estimates of Sustainable Agriculture.

Item	Estimate	Desv. Error	Wald	gl	Sig.	95% Confidence Interval	
						Lower Limit	Upper Limit
SA1	0.680	0.343	3.929	1	0.047	0.008	1.353
SA4	1.703	0.572	8.871	1	0.003	0.582	2.823

Note. Table 11 shows the degree of significance of the items with respect to the null hypothesis.

The table shows that both the acquisition of new sustainable technologies (SA1) and the development of sustainable agricultural resources (SA4) have a positive impact on agri-food production (b = 0.680 and b = 1.703, respectively), although with different levels of significance. In the case of SA1, its influence is marginally significant ($p = 0.047$) and its confidence interval reflects some uncertainty, suggesting that not all farmers have the same opportunities to access these technologies. On the other hand, SA4 shows a more noticeable and reliable impact, with a fully positive confidence interval ($p = 0.003$), highlighting that practices such as natural resource conservation and efficient water management have a more direct effect on strengthening agricultural sustainability.

4.3.3. Social Justice

The only estimate with a significance value of less than ($p < 0.05$) is the one concerning the quality of life of workers, with a significance level of ($p = 0.010$). This indicates that this variable has a relevant impact on agri-food production. Improving their quality of life not only represents an advance in terms of social justice, but also translates into productivity benefits. This implies that workers with better working conditions, access to basic resources, and greater stability are more motivated and committed to their performance, which directly contributes to positive results for the variable in question (see Table 12).

Table 12. Parameter estimates of Social Justice.

Parameter Estimates							
Item	Estimate	Error	Wald	gl	Sig.	95% Confidence Interval	
						Lower Limit	Upper Limit
SJ4	2.680	1.045	6.576	1	0.010	0.632	4.728

Note. Table 11 shows the degree of significance of the items with respect to the null hypothesis.

5. Discussion

This discussion addresses the findings obtained from the analysis of three key areas: market initiatives, sustainable agriculture, and social justice, within the context of the study conducted. Each of these aspects is examined in depth to evaluate their impact, limitations, and potential within the scope of the research. To enhance the accuracy of the results, the dimensions were calculated individually, filtering a total of 15 items, as a high number of questions may affect the consistency of Cronbach's alpha coefficient. Additionally, reflections on the main methodological weaknesses of the study are included, particularly concerning the use of convenience sampling, which may limit the generalisability of the findings. This approach seeks to provide a comprehensive overview that allows for an

understanding of the contributions of this research and the areas requiring greater attention in future studies.

5.1. About the Market Initiatives (H1)

The results of this study do not allow us to conclusively test the specific hypothesis 1, which states that international market incentives positively influence Peruvian agri-food production under Fairtrade schemes, motivating producers to improve the quality and sustainability of their products. Although the variables related to market incentives show an acceptable internal consistency ($\alpha = 0.751$), the lack of statistical significance in the analyses performed ($p > 0.05$) suggests that the proposed relationship is not conclusive. In particular, the relationships between bargaining power in prices (MI2, $b = 0.315$) and certification income (MI1, $b = 0.301$) do not reach significant levels, which limits the empirical validity of the proposed associations.

These limitations reflect the need to incorporate additional variables not considered in the model, such as organisational capacities, access to infrastructure and technology, and producer perceptions. These factors, critical for the effectiveness of incentives, could explain the complexity of the observed relationships, in line with previous studies, which highlight the structural and organisational barriers that condition the benefits of fair trade [19,75].

On the other hand, international market incentives play a relevant role in promoting the quality and sustainability of agri-food products in search of more lucrative markets. However, our findings indicate that, despite the reliability of the market incentive variables ($\alpha = 0.751$), the lack of statistical significance in the overall model ($p > 0.05$) highlights the need for further analysis of the structural and organisational elements that mediate this relationship.

Factor analysis provides clues to the underlying dynamics. Variables such as bargaining power in pricing (MI2) and certification income (MI1) are related to perceived economic benefits [13], while access to international customers (MI3) and competitiveness through certification (MI4) reflect broader impacts on global markets [10,14]. However, the lack of statistical significance in the ordinal logistic regression model ($p > 0.05$) questions the ability of these variables to capture the impact of incentives in the context of Fairtrade, as pointed out by Dragusanu et al. [19,75].

In addition, the study identifies key limitations, such as the absence of variables related to the organisational capacities of producers. Factors such as management training, access to market information, infrastructure, and modern technologies are essential to implement sustainable practices and improve competitiveness in international markets [17,24,39,58]. Dependence on intermediaries and value chains controlled by external actors, widely documented [19], also reduces producers' autonomy, limiting the direct benefits of incentives, as recommended in the literature [76].

Furthermore, producers' perceptions and motivations towards certification and incentives are determinants of their effectiveness. If the perceived costs outweigh the benefits, their commitment to these initiatives will be limited. Variables such as trust in certification systems, perceptions of equity in the distribution of benefits, and expectations of improved livelihoods need to be explored in future research [59,60].

Although Fairtrade certifications can foster sustainable practices and improve economic and social sustainability [17,24,39,58], the results of this study underline the importance of addressing structural, organisational, and contextual constraints. Future studies should adopt a multidimensional approach that combines qualitative (preferably ethnographic studies) and quantitative analyses to design inclusive strategies adapted to the needs of producers, maximising the benefits of Fairtrade in the Peruvian context.

A critical aspect that emerges from these findings is the potential mismatch between the theoretical foundations of Fairtrade incentives and their practical implementation in diverse socio-economic contexts. While the conceptual framework assumes that market incentives like certifications and improved bargaining power directly lead to enhanced quality and sustainability, this study demonstrates that such outcomes are far from guaranteed. The lack of statistical significance in key variables suggests that Fairtrade schemes often fail to account for the heterogeneity of producers' circumstances. For instance, small-scale producers in remote regions may face disproportionately higher costs to meet certification requirements, undermining the intended equitable distribution of benefits. Additionally, the overreliance on certification income as a primary metric of success neglects broader indicators of well-being, such as social cohesion, empowerment, and resilience to market shocks. This critique underscores the need for Fairtrade programmes to move beyond a one-size-fits-all approach, incorporating tailored interventions that address structural inequalities, improve access to resources, and build the organisational capacity of producers to fully leverage the opportunities provided by international markets.

5.2. About the Sustainable Agriculture (H2)

Our hypothesis H2, which stated that the relationship between sustainable agriculture and agri-food production mediated by Fairtrade is positive, was partially tested. The results show that the adoption of sustainable technologies (SA1) plays a significant role in the first group analysed (0.349), while the implementation of sustainable farming practices (SA3) and improved production methods (SA5) are distributed across groups 1, 3, and 5, respectively. This suggests that Fairtrade has facilitated the adoption of sustainable technologies in the companies studied. Furthermore, the data related to the parameter estimates of the ordinal regression indicate a positive and statistically significant impact on items SA3 and SA4 $(p < 0.05)$, supporting the effectiveness of the associated policies in enhancing agricultural sustainability. However, while Fairtrade's contribution to environmental sustainability is acknowledged, its role as a mediator in environmental conservation is questioned. According to Makita [39], in certain contexts, Fairtrade certifications have not fully achieved their purpose, underscoring the need for more detailed evaluations of their environmental effectiveness, as observed in this research.

These results support the notion that Fairtrade plays a key role in agricultural sustainability by facilitating access to innovative technologies and promoting responsible practices. They align with previous research [17,24,58–60], which highlights how these certifications contribute to strengthening the agricultural sector's resilience to global challenges. Nonetheless, we recognise that Fairtrade's contribution to environmental sustainability varies depending on the context, necessitating the implementation of specific strategies tailored to local characteristics, such as promoting agroecological practices and integrating targeted government policies.

From a theoretical perspective, our findings confirm that Peruvian agri-food Fairtrade can positively mediate between sustainable practices and agri-food outcomes, as largely determined by the literature [25,62,76–79]. However, we also identified that its effectiveness depends on external factors not considered in this study, such as government policies, access to financing, and local socio-economic conditions [62,80]. These external variables represent key areas that should be explored in future research to provide a more comprehensive understanding of the dynamics between Fairtrade certifications and agricultural sustainability.

Additionally, we believe that future studies should address how other external variables, such as international market dynamics, climate change, and consumer perceptions, moderate this relationship. For example, fluctuations in global demand and adverse cli-

matic conditions may limit farmers' ability to adopt sustainable technologies, even in the presence of certifications such as those offered by Fairtrade. Moreover, consumer behaviour and their willingness to pay for certified products are critical factors in ensuring the long-term economic viability and sustainability of these practices.

In conclusion, our hypothesis was generally supported but with significant nuances that highlight areas for improvement and opportunities for future research. We acknowledge the importance of adopting a comprehensive approach that considers both the current limitations of Fairtrade and the impact of external variables. Such an approach would enable the design of more effective strategies to strengthen the economic, social, and environmental sustainability of the agri-food sector in diverse contexts, consolidating its resilience to global challenges.

While the data strongly support the positive impact of Fairtrade certifications, the findings also reveal critical limitations in their scope and application, particularly due to the convenience sampling used in the survey. The significant variation in the adoption and effectiveness of sustainable practices underscores the need for a more nuanced understanding of how Fairtrade interacts with contextual variables. For instance, discrepancies in the results across different groups indicate potential gaps in the design or implementation of Fairtrade policies, especially when addressing region-specific challenges. Moreover, the emphasis on technological adoption and farming practices might overshadow other essential dimensions of sustainability, such as social equity and economic resilience, which are equally vital for achieving comprehensive sustainability outcomes. These observations highlight the importance of refining Fairtrade frameworks to enhance their adaptability and effectiveness in diverse agricultural contexts.

5.3. About the Social Justice (H3)

The findings of our study partially support the proposed hypothesis. Regarding the organisations that participated in our survey, the results suggest that while progress has been made in certain aspects of social justice, the full achievement of this goal remains elusive. We observed that Fairtrade has significantly influenced the distribution of benefits and the improvement of working conditions within Peruvian agri-food value chains. Specifically, the prevention of child labour (SJ1), the commitment to sustainable agriculture (SJ2), and the improvement of workers' quality of life (SJ4) demonstrated significant loadings in the ordinal regression analysis ($p < 0.05$). However, the inclusion of marginalised groups (SJ3), although noteworthy, was highlighted by the positive impact of the parameter estimates in the ordinal regression ($p = 0.010$).

We recognise that, although the Peruvian Fairtrade model has proven effective in promoting social justice and empowering certain groups, as widely suggested by the literature, the benefits do not always reach the most vulnerable actors in value chains, as noted by Mook et al. [4,28], Barro et al. [1], and Guevara-Rosero et al. [10]. Our findings align with these observations, confirming that Fairtrade has increased producers' incomes, yet significant challenges remain concerning the inclusion of marginalised communities, as confirmed by previous studies [1,24]. This scenario raises questions about whether current certification mechanisms are sufficiently inclusive and sustainable to address these limitations. Consequently, this highlights that the organisations participating in the survey still face barriers to ensuring comprehensive social justice, particularly in terms of inclusion and equity within value chains. These findings underscore the need to review and strengthen both internal and external mechanisms driving Fairtrade, ensuring that benefits reach all actors, especially the most vulnerable [1,14,24].

Additionally, we believe that one of the areas not addressed in our study, but crucial to understanding these dynamics, is the role of power within the cooperatives and

associations involved. Previous research, such as that by Raynolds et al. [54], suggests that democratic decisions can be influenced by actors with greater power, limiting the equitable distribution of benefits. Moreover, it is necessary to assess how social premiums are allocated to community projects and whether these investments truly reach those most in need. Incorporating participatory monitoring systems could be an effective strategy to address these gaps.

Similarly, we identified the need to explore how changes in the global market affect the effectiveness of Fairtrade. Factors such as international price fluctuations and government policies could influence the sustainability of Fairtrade standards in key products like coffee and cocoa [55,56]. Our study did not analyse these aspects in depth, but we believe their inclusion in future research could provide a more comprehensive view of the limitations and opportunities of Fairtrade in the Peruvian context.

In conclusion, we confirm that Fairtrade makes a significant contribution to social justice in Peruvian agri-food production. However, important challenges persist, such as the inclusion of marginalised groups and the equitable distribution of income. We believe future research should focus on analysing power dynamics within value chains, the impact of global market fluctuations, and the influence of public policies. This would enable the development of more inclusive and sustainable strategies, ensuring that the benefits of Fairtrade reach all actors involved [4,54,57].

These findings also raise methodological issues that warrant critical reflection. Although a robust approach was employed using ordinal regression analysis, the study design, due to convenience sampling, did not allow for the capture of power dynamics within organisations or the internal processes that may constrain the equitable distribution of benefits. For instance, previous research has indicated that leadership concentrated in actors with greater economic or political influence within cooperatives can restrict opportunities for more vulnerable groups. This methodological gap suggests that a more interdisciplinary approach, combining qualitative tools such as in-depth interviews with longitudinal quantitative analyses, could provide a more comprehensive and explanatory perspective on the observed phenomena.

Moreover, it is essential to consider the impact of exogenous factors, such as fluctuations in international prices and government policies, which were not thoroughly explored in this study. These external variables can significantly influence the sustainability of Fairtrade standards, particularly for key products like coffee and cocoa. The lack of analysis in this area limits the generalisability of the results and the study's ability to offer strategic recommendations. Incorporating these elements into future research, through approaches such as economic scenario modelling or policy evaluation, could broaden our understanding of the structural challenges faced by Fairtrade and contribute to the development of more inclusive and resilient strategies for actors within Peruvian agri-food value chains.

5.4. Limitations

The analysis of the impact of Fairtrade on Peruvian agri-food production reveals various methodological and contextual limitations that affect the validity and generalisability of the results. Although the convenience sampling employed in this study has practical justifications, it is essential to acknowledge that this methodology is inherently linked to significant biases. Relying on data collected from cooperative leaders and companies participating in events such as Expoalimentaria 2024, the findings may be more influenced by individual perceptions and the specific dynamics of the event rather than an objective and balanced representation of reality, as indicated in the literature [81].

This approach, while useful in contexts where participant access is restricted by time or resources, compromises sample diversity and the ability to generalise the findings to

other regions or communities. For instance, the specific setting of the data collection may lead to an overrepresentation of actors with direct commercial interests in Fairtrade, excluding smaller producers or cooperatives with less exposure to international markets. This introduces a critical limitation: the data obtained may reflect the priorities and experiences of a specific subset of actors rather than the broader dynamics of Fairtrade in the country.

Despite these limitations, certain aspects of the study's design strengthen the representativeness of the data collected. The geographical segmentation analysed in Table 8 of the surveyed companies aligns with the national proportions of Fairtrade certifications. For example, the central and northern regions, which lead national certification rates, are significantly represented in the sample. Similarly, although the southern and jungle regions have lower participation; their specialisation in products such as coffee and cocoa reflects their contribution to the Fairtrade ecosystem, as empirically detailed in secondary evidence from governmental organisations such as PromPerú [66].

Another aspect reinforcing the study's validity is the correspondence between Fairtrade-certified products and regional specialisations. The central region, for instance, excels in exporting cocoa, coffee, and quinoa, while the north focuses on high-demand products such as bananas and mangoes. These similarities between the survey findings and national data partially mitigate biases introduced by non-random sampling, providing a more reliable basis for interpreting the results, as suggested by methodological literature [82].

However, the inherent limitations of convenience sampling cannot be overlooked. Relying on events such as Expoalimentaria 2024 for data collection introduces risks of selection bias, as participants in such forums tend to be better-organised companies and cooperatives that have participated in previous editions of Peru's main agri-food export fair [68,69]. This excludes actors who did not attend the 2024 edition but are also part of the Fairtrade ecosystem, restricting the diversity and depth of the analysis. Additionally, the event setting may influence participants' responses, which could be rushed or conditioned by the commercial context of the forum.

From a thematic perspective, measuring the long-term effects of Fairtrade poses another significant challenge. Benefits such as sustainability, poverty reduction, and rural development require longitudinal studies to capture their full impacts. Short-term research, such as this study, may not identify deep structural changes, highlighting the need for more comprehensive and long-term designs with geographically representative scope.

External factors such as government policies, international prices, and climatic conditions also influence agri-food production, complicating the direct attribution of observed benefits to Fairtrade. These elements underscore the importance of incorporating stricter controls in future studies to avoid misinterpretations.

Methodologically, this study highlights the need to incorporate mixed approaches that combine representative quantitative analyses with in-depth qualitative studies. Such methodologies would allow a more comprehensive assessment of critical variables such as agricultural practices, social dynamics, and economic factors, providing a more holistic view of Fairtrade's impacts.

Despite these limitations, the findings are significant and provide empirical evidence on how international market incentives influence the adoption of sustainable practices. While some key variables did not reach statistical significance, the results offer valuable starting points for future research and underscore the importance of addressing structural and contextual inequalities in agricultural communities.

The study also highlights the relevance of adopting more inclusive sampling designs that better represent different regions and crop types in Peru. This would not only improve

the generalisability of the results but also facilitate the development of more effective strategies to maximise the benefits of Fairtrade in specific agricultural areas.

Moreover, it is crucial to address producers' perceptions of the equity and fairness of Fairtrade, as these opinions can significantly influence data collection and interpretation. Cooperative leaders and company representatives at events like Expoalimentaria 2024 may have provided responses conditioned by their experiences or expectations, introducing a subjective margin that should be considered in the analysis. Future research would benefit from complementing surveys with in-depth interviews or focus groups, enabling a richer and more nuanced understanding of the perceptions and motivations of the actors involved.

Finally, this study emphasises the need to evaluate Fairtrade impacts beyond traditional economic indicators. Factors such as the well-being of agricultural communities, environmental conservation, and resilience to climate change are critical areas that should be included in comprehensive analyses. Incorporating a multidimensional approach that evaluates economic, social, and environmental benefits would strengthen the understanding of Fairtrade as a driver of sustainable development, providing a solid foundation for designing public policies and business strategies that promote more inclusive and sustainable agricultural practices.

In conclusion, this analysis underscores the importance of a rigorous and contextualised approach to studying the impact of Fairtrade on Peruvian agri-food production. The limitations identified represent opportunities for improving future research, ensuring it captures the complexity and diversity of Fairtrade in different regional and economic contexts. With a more robust methodological design and an integration of mixed approaches, it will be possible to advance the promotion of inclusive strategies that maximise the benefits of Fairtrade for agricultural communities in the country.

6. Conclusions

We conclude that Peruvian Fairtrade has a significant but uneven impact on various dimensions of the country's agri-food production. While this study provides valuable insights into international market initiatives in the context of Fairtrade in Peru, our findings do not fully confirm the hypothesis regarding its positive influence on the quality and sustainability of agri-food production. Although the analysed variables demonstrate acceptable internal consistency, the lack of statistical significance highlights the complexity and limitations of the proposed model.

We consider it crucial to incorporate additional variables, such as organisational capabilities, access to infrastructure and technology, and producers' perceptions. These factors are essential to better understand the relationship between market incentives and expected benefits. The results underscore the need to design more inclusive strategies tailored to the diverse realities of producers, aiming to overcome structural and organisational barriers that currently limit the potential benefits of Fairtrade.

Additionally, we have identified that market incentives face challenges in achieving effective implementation due to varied socio-economic contexts, the high costs associated with certifications, and dependency on intermediaries. This underscores the need for a multidimensional approach that goes beyond certification as the primary metric of success. Instead, we propose integrating indicators such as social cohesion, empowerment, and resilience to market dynamics. We believe Fairtrade programmes must evolve from standardised models towards specific interventions that empower producers, enhance their access to resources, and strengthen their organisational capacities, thereby maximising opportunities in international markets.

Furthermore, we observe that the relationship between sustainable agriculture and agri-food production, mediated by Fairtrade, yields promising results, albeit conditioned by various contextual factors. While the data support the positive influence of Fairtrade certifications on adopting sustainable technologies and responsible practices, they also reveal significant limitations in their scope and application.

These findings lead us to emphasise the need for strategies that incorporate external factors such as government policies, access to finance, and local socio-economic conditions. Addressing these factors, alongside international market dynamics and the effects of climate change, is fundamental. They not only moderate the relationship between Fairtrade and sustainability but also influence farmers' ability to effectively adopt sustainable technologies.

Overall, we confirm that Peruvian Fairtrade can play a key role in promoting agricultural sustainability. However, we believe significant adjustments are required in regulatory frameworks and approaches tailored to regional characteristics. A comprehensive strategy addressing these limitations can strengthen the economic, social, and environmental sustainability of the agri-food sector, improving its resilience to global challenges.

Finally, we affirm that Fairtrade plays a significant role in promoting social justice within Peruvian agri-food value chains, achieving improvements in labour conditions and the quality of life for many stakeholders. Nevertheless, challenges persist, such as the effective inclusion of marginalised groups and the equitable distribution of income. We consider that current certification mechanisms, while effective in certain aspects, must be strengthened to ensure that benefits reach the most vulnerable actors in these chains.

7. Future Research

The present study on the impact of Fairtrade on Peruvian agri-food production has identified methodological and contextual limitations that pose challenges but also offers opportunities for more innovative future research. While convenience sampling enabled access to information within a restrictive environment, it is necessary to explore more robust approaches that mitigate the inherent biases of this technique, thereby facilitating the collection of representative and generalisable data.

In this context, it is essential to incorporate sampling methods that prioritise geographical diversity and the inclusion of less visible actors within agri-food value chains, such as small-scale producers and local cooperatives. Designs like stratified random sampling, alongside data triangulation techniques, could enhance the analysis and provide a more accurate understanding of the dynamics of Fairtrade across different regions of Peru. This would not only promote equity in data representation but also allow for the identification of specific patterns to inform tailored strategies according to local characteristics.

Furthermore, capturing the long-term effects of Fairtrade on structural indicators such as sustainability, inequality reduction, and rural development necessitates adopting longitudinal approaches. These approaches can combine historical data analysis with contemporary tools such as simulation models and predictive analytics to examine how Fairtrade impacts the evolution of producers' incomes, their access to international markets, and their resilience to global economic fluctuations.

Thus, advancing towards mixed-method methodologies is crucial to understanding the interplay between economic, social, and cultural variables. The integration of rigorous quantitative analyses with in-depth qualitative studies, such as semi-structured interviews, focus groups, and ethnographic analyses, will enable the exploration of intangible dimensions such as perceptions of social justice, producer satisfaction, and adaptability to new international regulations. These perspectives can be complemented by impact assessments that precisely measure Fairtrade's transformative capacity within producer communities,

ensuring that future research is not only comprehensive but also relevant and applicable to the design of more inclusive public policies and market strategies.

As a recommendation for future research, it is essential to explore strategies for reducing certification costs in Fairtrade systems, aiming to make them more accessible to small-scale producers. Further studies should focus on designing and evaluating financial support mechanisms, such as subsidies, microloans, or shared funding models, to enable producers to participate in Fairtrade initiatives without compromising their profitability. These efforts would contribute to fostering inclusivity and sustainability in agri-food production.

In addition, it is important to explore further the implementation of technologies that can connect farmers directly to international markets, minimising dependence on intermediaries. Future work should focus on developing systems that provide detailed and consistent information, even in regions with poor infrastructure, to obtain much more accurate and representative results.

It is also important to carry out research that considers geographical and cultural differences, exploring how the specific conditions of each area influence the benefits obtained by small producers. This should include direct visits to specific regions to observe and analyse in more detail the influence of Fairtrade in local contexts.

To optimise the impact of Fairtrade on Peruvian agri-food production, it is crucial to develop specific lines of research with clear implementation plans. This approach will ensure representative and actionable results, which are fundamental for designing effective public policies and trade strategies.

First, reducing certification costs, a significant obstacle for small producers, requires priority attention. Comparative studies are proposed that analyse certified and non-certified cooperatives in terms of operational costs, financial benefits, and long-term sustainability. Inspired by the methodological design of Ruben and Fort [24], this study would include financial records of local cooperatives, interviews with agricultural leaders, and analysis of government subsidies. To ensure the reliability of the results, we suggest employing data triangulation, combining structured surveys, official records, and direct testimonies from producers. This approach will identify specific areas where intervention could reduce costs without compromising quality or certification standards.

Secondly, it is essential to explore the impact of digital technologies in connecting small producers directly to international markets, minimising dependence on intermediaries. We suggest developing a pilot project in the central macro-region of Peru, which concentrates the largest number of certified companies, where the effect of digital direct trade platforms on producers' income and operating costs could be assessed. Based on the approach of Dragusanu et al. [75], the study could employ longitudinal methods to measure changes in trade dynamics over a period of 2–5 years. Data sources would include pre- and post-implementation surveys, sales records on digital platforms, and interviews with international producers and buyers. This methodological design would therefore allow the assessment of not only technological adoption, but also the associated tangible economic benefits.

Third, based on the recommendations of Dietz et al. [6], we suggest conducting studies that focus on the evolution of income, productivity, and participation in international markets. For this purpose, historical records, market studies, and interviews with key actors in the sector will be used. The application of econometric techniques will allow for controlling external variables, such as government policies and global market fluctuations, thus ensuring a rigorous analysis of the net impact of certifications.

Finally, the structural and organisational factors of Fairtrade play a pivotal role in shaping the relationship between this certification scheme and agri-food production. Ele-

ments such as cooperative governance, resource allocation, and the capacity for collective bargaining directly influence the efficiency and sustainability of Fairtrade initiatives. These factors determine how producers access benefits like price premiums, technical support, and market opportunities. Moreover, they affect producers' ability to implement sustainable practices and maintain the quality required by international markets. Future research should focus on examining how these organisational dynamics mediate the relationship between Fairtrade and the production processes of agri-food systems, particularly in contexts with significant socio-economic inequalities. A deeper understanding of these factors could inform the design of more robust frameworks that enhance the inclusivity and effectiveness of Fairtrade models.

Equally important is the exploration of how structural barriers—such as limited access to infrastructure, technology, and education—hinder the potential of Fairtrade to support small-scale producers. These barriers often perpetuate dependency on intermediaries and restrict producers' ability to fully engage in value chains. Investigating these aspects in greater depth would enable researchers to identify strategies to overcome these challenges, such as fostering partnerships between public institutions, private stakeholders, and producer organisations. Additionally, longitudinal studies could reveal how improvements in these structural conditions impact the long-term success of Fairtrade in promoting sustainable agri-food production. By prioritizing these areas, future investigations can contribute to strengthening the alignment between Fairtrade principles and the realities faced by producers, ensuring a more equitable and sustainable agri-food sector.

Author Contributions: Conceptualization, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L., C.A.A.M. and J.R.M.C.; methodology, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L., C.A.A.M. and J.R.M.C.; validation, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L. and C.A.A.M.; formal analysis, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L., C.A.A.M. and J.R.M.C.; investigation, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P. and T.A.L.; resources, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L., C.A.A.M. and J.R.M.C.; data curation, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L. and C.A.A.M.; writing—original draft preparation, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L., C.A.A.M. and J.R.M.C.; writing—review and editing, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L., C.A.A.M. and J.R.M.C.; visualization, D.L.L.P., H.O.S., K.N.d.P.Y.D., M.E.D.C., V.L.S.S., J.P.P., T.A.L. and J.R.M.C.; supervision, J.R.M.C.; project administration, J.R.M.C.; funding acquisition, J.R.M.C. All authors have read and agreed to the published version of the manuscript.

Funding: The authors would like to acknowledge Dirección de Investigación de la Universidad Peruana de Ciencias Aplicadas, which provided financial support through UPC-EXPOST.

Institutional Review Board Statement: Ethics Committee of the Research Group on Management and International Business at UPC approval number 04-2024-GININT.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: No new data were created or analysed in this study. Data sharing is not applicable to this article.

Acknowledgments: The authors appreciate the support received from the research direction of the Universidad Peruana de Ciencias Aplicadas.

Conflicts of Interest: The authors declare no conflict of interest.

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